

Dawnwood Interactive

The complete integer kernel

Literal-source formalization, numerical evidence and ELI5 performance metrics

Model **integer-1** | report version 0.4 | snapshot version 4
16 September 2026 | Project author: Tom Klootwijk

Delivered definition / measured evidence	Result and scope
Quantum-style mechanisms	Actual integer two-path Hadamard interference and H followed by H are probed. Raw H-squared minimum normalized fidelity is 0.999985; LP8 between gates materially changes this result.
Whole recurrence	Integer geometry, chart motion, phase calculus, Hadamard, encoding, feedback and code-word mutation. The native sampler has an explicit typed-interface boundary.
Compact active hot data	4,004 bytes : operator words, small log-polar/trigonometric LUT and live routing masks. This excludes machine code, driver resources and staging.
Exact cancellation and seams	All 256 symbolic antipodes cancel exactly; 36,864 integer Klein seam cases have zero mismatches. These are component diagnostics.
Saved throughput example	cached / bc5: 7,216.47 million pair updates/s . Workload and comparison scope appear on page 10.
Physical interpretation	A source-inspired classical digital automaton. LP8 radius wrapping remains nonunitary; physical double-slit behavior is not claimed.

What changed from the earlier edition

This edition implements the complete numerical circulation with integer words. Signed SDF calculations are part of it; they are not the entire change. Exact symmetric LUT construction removes floating cancellation residue. The fourth-slot two-unit Y event, moving operators, B/A state and self-reference remain explicit parts of the law.

The optional 64 KiB D4 pair cache reuses exact finite integer results. The measured cached variant is the recommended executable; the 4004-byte compact variants remain available for comparison. Full snapshot checks precede interpretation of the speed ratios.

Source definitions, chosen numerical bindings, CPU component diagnostics, GPU state checks, instruction audits and hardware performance are separate evidence classes throughout this report.

The literal document, component by component

Page references below refer to the supplied 24-page double-slit-theory.pdf. The original user requirements are distinguished from the other assistant's interpretations and unsupported predictions.

Source / requirement	Integer realization and explicit boundary
3-4 / two pinions, Hadamard, log-polar LUT	Two complete complex token streams. Q11 integer sum/difference with 23170/32768 scaling and specified rounding.
4-6 / parity, one-bit jitter, [0,2,0,1]	Packed population parity and deterministic one-bit jitter. First-block tokens R=0, G=176; B=0 and A=0 encode history and inverse identity.
7 / two Y-up events in fourth RK slot	A +2 Q16 phase-unit event enters only the fourth derivative argument. Integer rounding is defined at each stage.
8-9 / shapes, double-dot, phase difference	Signed Q18 circle, T, triangle, cone-meridian, ambient sphere and apex supports; Q22 dot product; wrapped second phase difference.
8-9 / Fibonacci phyllotaxis and blend	Integer log-radius placement, Q16 golden-angle step 25032, and executable fixed-word blend instructions. No unmeasured uniform-area claim.
9-12 / dichromatic RG, B, inverse T in A	R and G each encode a full stream. Exact per-block B/A words. A inverts two modular phase words, not the whole lossy update.
12-14 / SDF Klein surface, no raster grid	Integer quotient coordinates and a four-coordinate K realization attach moving signed supports to the surface. Arrays supply storage, not geometry.
14-16 / operators themselves change at Psi	Field feedback rewrites body words, links, support, coefficients, placement and routing. The next interval consumes the changed data.
18 / small cached substrate, resident packed field	1540-byte math LUT plus 1984-byte operator bank and 480-byte live masks. GPU-owned packed pages, ordered commits and bounded staging.
20-22 / packing, complexity, implicit tree	Real byte counts and full-sweep work; implicit children $2i+1$ and $2i+2$. Four decisions choose an operator leaf; no billion-node payload search is implied.
12 / optional Bayer readout	Downstream and optional. No renderer participates in recurrence or is needed for these checks.
16-23 / physics, optimality and universality	These require separate definitions, experiments or proofs. Mutability alone supplies neither a reward objective nor quantum behavior.

Phi ambiguity resolved: the user says log-encoded polar radius PHI, but gives no logarithm-of-phase definition. This binding uses logarithmic radius and linear modular phase. A logarithm of phase would require another branch/zero/periodicity convention; it is not silently invented.

Integer words and the small LUT

Quantity	Exact representation / chosen units
LP8 token	8 bits: high nibble 1..15 is log-radius cell; low nibble is one of 16 phase cells. All 16 high-zero patterns decode zero.
Chart and phase	Q16: 65,536 units per normalized log-radius period or phase turn. Extended signed u preserves winding before fold.
Complex vector	Two signed Q11 components packed into uint32; scale 2048. Antipodal entries are exact negations. Products use widened intermediates.
Sine lookup	257 unsigned Q15 quarter-wave values; scale 32,768, integer interpolation, exact quadrant symmetry.
Signed geometry	Q18 distance/support values; scale 262,144. Radius word is low16(support)+1.
Config	dt_q16=1024 (1/64), inverse_gain_q16=16384 (1/4). Allowed dt: 1..65536; gain: 0..65536.
Math LUT payload	256 uint32 LP words + 257 uint16 sine words = 1538 bytes; aligned sizeof(MathLut)=1540 bytes.

Encode, decode and fold

```
h=q>>4; p=q&15; h=0 means exact complex zero
u=(h-1/2)/15; v=(p+1/2)/16 [real definition]
z=2^(-4+7.5*u) * exp(i*2*pi*v) [rounded to Q11]

Uword=round((log2_q16(x*x+y*y)-14*65536)/15)+du
Vword=atan2_turn16(y,x)+dv
(u+65536,v) ~ (u,-v); (u,v+65536) ~ (u,v)
```

The runtime encoder uses integer repeated-squaring log2 and integer CORDIC phase. These supply finite Q16 words; the real equation specifies what the LUT samples mean. Zero is tested before logarithm or offsets. Negative radial winding uses floor semantics, so odd winding reverses phase consistently.

Rounding is symmetric nearest with ties away from zero where specified. Chart/phase wrap; supported products use bounded or widened integers. Fixed-point words are not exact real numbers. No arbitrary epsilon cutoff is introduced.

One complete integer interval

State F contains the packed field P, 31 operator records O, exact per-block B/A, epoch e, cursor c and seed. W blocks form one update window; changing W changes how much field evolves before the operators mutate.

```
i(t)=(c+t) mod N, 0<=t<W
staged[t]=E(P,O,B,A,i(t),e,seed)
commit staged results to distinct blocks
mutate next operator bank from old O and staged feedback
rebuild live integer routing masks; advance c and e
```

Data-selected page hops supply neighbours. Four implicit-tree decisions select a leaf 15..30. Its first referenced body, second referenced body and own body execute on eight 16-bit registers; overflow is modular. Body-reference 31 aliases root 0.

```
drive=(round(s16(reg0)/2), round(s16(reg1)/2)+25032)
k=round((low16(coeff)+1)/16) [all Q16]
f(q)=(drive.u + round(k*sinQ15(q.v)/32768),
      drive.v + round(k*cosQ15(q.u)/32768))
a=f(q)
b=f(q + round(dt*a/131072))
c=f(q + round(dt*b/131072))
d=f(q + round(dt*c/65536) + (0,2))
next=q + round(dt*(a+2*b+2*c+d)/393216)
```

The +2 event is faithful to source page 7. This is a rounded forced four-stage map; textbook fourth-order real-ODE accuracy is not claimed. Phase displacement adds dt times the wrapped neighbour second difference, then the previous inverse words feed back through the integer gain.

```
Hplus=round((zR+zG)*23170/32768)
Hminus=round((zR-zG)*23170/32768)
twist=round(dt*(xR*xG+yR*yG)/4194304)
Rnext=Q(Hplus, du, qa+twist) XOR jitter
Gnext=Q(Hminus, -du, qb-twist) XOR jitter
```

A stores -pa and -pb modulo 65,536. It does not invert dot twist, Hadamard, radius fold, codec loss or mutation. B rolls history with event/return/output parity. Feedback also changes code bits, references, signed support, coefficients and positions. Stored interpreter words change; the CUDA ISA does not rewrite itself.

Signed SDFs, surface attachment and masks

```
K(u,v)=((2+.5*cos(2*pi*v))*cos(2*pi*u),
        (2+.5*cos(2*pi*v))*sin(2*pi*u),
        .5*sin(2*pi*v)*cos(pi*u),
        .5*sin(2*pi*v)*sin(pi*u))
K(u+1,-v)=K(u,v)
```

This real parameterization defines the geometry sampled by integer Q18 operations. Canonical integer folding makes quotient-equivalent inputs produce identical integer coordinates. It attaches operator locations to K by construction, without ray traversal or projection iteration.

Support	Integer implementation and its limits
Circle / apex	Nearest integer radial distance; subtract radius for the circle. Apex is a point-distance field.
T	Minimum of two box signed fields. Membership is meaningful; the union is not exact Euclidean interior distance throughout overlaps.
Pyramid side / cone	Signed triangle distance from projected edges; cone means the explicitly chosen two-dimensional meridian.
Sphere	Radial support from ambient four-coordinate K distance, restricted to the chart surface.
Near a boundary	Circle/sphere/triangle preserve integer inside/outside membership with -1 or +1 when distance rounding would create a false zero.

A 480-byte cache of live decisions

```
15 internal nodes * 256 LP8 inputs * 1 bit = 480 bytes
mask[node,token] = (signed_support(node,token_chart)>=0)
```

The full signed integer support functions remain executable and define these bits. The optimized rebuild uses exact integer sign predicates that avoid unneeded square roots and projections; comparison against signed evaluation checks this specialization. Masks are rebuilt from every current operator bank. Jitter, history and route parity still combine with the cached predicate at runtime.

The sign specialization passed 187,164 CPU comparisons and 3,598 boundary/coverage assertions with zero failures. GPU checks compare all 3,840 live support bits with the signed CPU reference.

The source gives no global scalar signed-distance equation for a Klein bottle. This binding uses quotient-constrained locations and signed local supports. It does not assert a global inside/outside field for a non-orientable surface, or replace all geometric values with Boolean masks.

Cache footprint and the native sampler boundary

Active data component	Compact	Optional exact D4 cache
31 operator records	1984 B	1984 B
Integer math LUT	1540 B	1540 B
15 x 256 routing decisions	480 B	480 B
Canonical pair cache	0 B	8192 x 8 B = 65,536 B
Logical active data total	4004 B	69,540 B

These totals exclude CUDA machine code, interpreter implementation, stack/register spills, the second operator bank, allocation alignment, handles, driver/runtime resources, staging and payload. Exact allocated/shared bytes belong in each run report. The small representation is not a claim that the entire program fits in 4004 bytes.

The optional D4 table stores canonical integer Hadamard/log/phase outputs under four quarter-turn rotations and reflection. The minus output reuses plus with an antipodal second input. A record stores u, phase and orientation correction bits. CPU cache checks passed 1,253,890 assertions; the GPU cache check exactly matches both output charts across all 65,536 pairs. Full snapshot equality is checked separately.

Integer data arithmetic is narrower than zero floating opcodes

Native BC5 hardware decoding retains a float-typed CUDA sampler interface. Centre-coordinate bits are built with integer operations and returned UNORM bits are recovered as integer tokens. RG8 returns integer texels. Bit-preserving moves and this explicit fixed-function boundary are reported, rather than hidden.

Machine-code exception: SASS uses HFMA2 with architectural zero-register operands and literal constants to materialize fixed bits. The verifier permits only the recognized constant-only form. This is a floating opcode, so **zero floating opcodes would be false**. It does not perform input-dependent floating arithmetic. Dynamic floating arithmetic/conversions/comparisons and SFU reciprocals remain prohibited; both PTX and SASS must be audited.

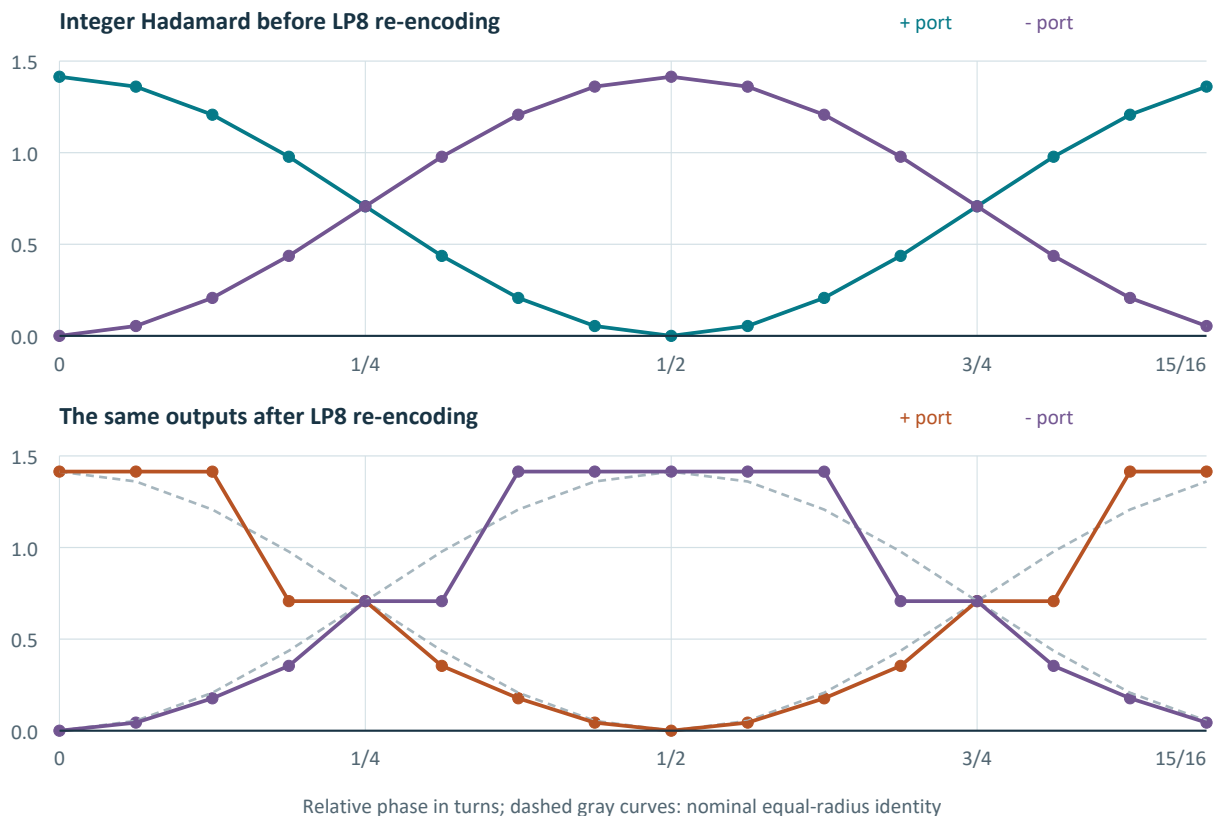
Saved instruction audit	Result / scope
Actual executable audit	All four shipped executables pass PTX and SASS checks; binary SHA-256 matched before summary.
Permitted boundaries	Native f32 sampler/bit transfers and strictly constant-only HFMA2 materialization are listed separately. No data-dependent floating arithmetic/conversion/SFU found.
Audit record	results/integer/instructions/build_audits.json; no zero-floating-opcode or driver-internals claim.

Cache preference and persistence are hardware policies, not locks. A larger exact table may win or lose depending on reuse and occupancy; its inclusion does not itself establish a speedup.

Two-stream interference: measured component behavior

```
a=r*exp(i*theta); b=r*exp(i*(theta+phi))
Iplus=r*r*(1+cos(phi)); Iminus=r*r*(1-cos(phi))
Iplus+Iminus=2*r*r [isolated real Hadamard]
```

The diagnostic sweeps sixteen phase differences at radial nibble 8. Solid sample curves use the actual integer functions; dashed curves use the nominal equal-radius identity. Connecting samples guides the eye and does not introduce a physical screen or propagation model.



Exact symbolic cancellation: 0 failures across 256 symbols, including sixteen zero aliases and 240 nonzero values. Equal-input difference cancellation and encoded antipodal zero also have 0 failures.

The rounded integer input vectors have slightly phase-dependent radii. The JSON therefore also records the exact real sum/difference identity for those actual decoded vectors, separate from the nominal curve. Raw integer port energy differs from that identity by at most 0.0004480 in this sweep.

This validates finite interference algebra and its error. The next page tests quantum-style two-arm compositions using the production math. The full field recurrence has no detector, Born measurement, entangled tensor-product register or established quantum-state fidelity.

Quantum mechanisms: fidelity, fringes and lost norm

Two probes exercise the production integer primitives: apply H twice, and split one occupied arm with H, rotate the second arm, then recombine with H. These are isolated CPU mathematical circuits, not a substitute simulation of the full evolving field.

```
ideal H*H = identity
ideal Pminus after H-phase-H = sin(phi/2)^2
F = |<input,output>|^2 / (norm(input)*norm(output))
R = norm(output)/norm(input)      [squared vector norms]
```

F compares state direction up to a global phase and is normalized only in the diagnostic. R separately reveals amplification or loss. The kernel does not normalize its state or perform measurement. Reported probabilities are host readouts, not implemented Born-rule events.

Recovery over 65,280 nonzero pairs	Minimum F	Median F	Norm ratio R range
Integer H then H	0.999985	1.000000	0.987143 to 1.007730
H / LP8 / H / LP8	0.016916	0.997436	0.020832 to 8,216.13

The census includes all 65,536 token pairs and excludes 256 zero/zero alias pairs from normalization. A high median fidelity cannot certify every state: LP8 wrapping produces a worst-case direction mismatch as well as large norm changes.

Fringe-shape accuracy, not just a bright and a dark point

Probe / radial bin	Raw max error, percentage points	LP8 max error, percentage points
Equal arms / 1	0.43236	96.09653
Equal arms / 9	0.00692	3.53707
Equal arms / 15	0.00200	96.09670
H-phase-H / 1	0.33610	96.09669
H-phase-H / 9	0.00750	3.53707
H-phase-H / 15	0.00365	3.53461

Each row uses sixteen phase settings. All these sampled port-energy visibilities are 1, using (maximum-minimum)/(maximum+minimum), because exact dark ports survive. Yet some encoded fringe shapes deviate by about 96 percentage points. Visibility alone conceals that failure.

Application evidence: the raw integer H and phase operations are useful two-path interference building blocks at measured precision. The present LP8 quotient is not a general quantum-state codec. Full nonlinear motion, history/inverse feedback, BC5 loss and operator mutation have not been shown to implement unitary quantum evolution.

Numerical accuracy and limits of the quotient

Executed diagnostic	Measured result / interpretation
Hadamard before LP8	Median relative norm error 0.0113%; maximum 0.7316%. All 65,536 pairs checked; 256 zero/zero pairs excluded from ratios.
After LP8 re-encoding	Energy ratios range from 0.00003042 to 32,867.815. Median absolute relative energy error 1.5433%; 95th percentile 41.0609%.
Concrete wrapping witness	Input tokens [0,16] produce [255,247] after the encoded split. Exact cancellation does not stop small nonzero radii from wrapping into the high-radius cells.
In-window amplitude quantization	Median 8.6320%; 95th percentile 16.7266%; maximum 19.3762%. 245,744 accepted samples from the stated radius/phase grid.
Source amplitude 2	Token 176 decodes magnitude 2.37819682, phase 11.24764 degrees; relative amplitude error 18.9098%.

Radius folding identifies magnitudes differing by $2^{7.5}$, approximately 181.02, with phase reversal. This computational quotient does not conserve ordinary wave energy. RG8 removes BC5 token-byte loss; it does not remove LP8 quantization or the quotient.

Exact invariants and the fourth-stage event

Check	Result
Klein coordinate seam	36864 cases; 0 integer coordinate mismatches.
Modular inverse phase	65536 words; 0 composition failures.
Forced four-stage map	16384 cases; 0 mismatches against independently assembled integer stages. The event changes 271 final outputs; rounding hides its effect in the others.

The amplitude sample uses 960 logarithmic radial midpoints by 256 phase midpoints, quantized to Q11. Sixteen rounded inputs outside the actual nominal radius window are excluded and counted. These deterministic populations are diagnostic scopes, not measured application distributions or statistical confidence intervals.

Performance: how fast the items are revised

One token pair is one two-stream item. Pair updates per second count revisions, not unique stored items, physical particles, frames, FLOPS or solved tasks. A fair optimization comparison holds the integer numerical law and recurrence parameters fixed.

Variant / codec	Cache	Million updates/s	Wall ms	Reference ratio
reference / bc5	balanced	2,033.04	132.04	1.00
shared / bc5	balanced	459.80	583.80	0.23
texture / bc5	balanced	460.57	582.83	0.23
cached / bc5	balanced	6,856.50	39.15	3.37
cached / bc5	l1	7,216.47	37.20	3.55
reference / rg8	balanced	2,334.60	114.98	1.00
shared / rg8	balanced	534.92	501.82	0.23
texture / rg8	balanced	535.86	500.94	0.23
cached / rg8	balanced	5,863.79	45.78	2.51
cached / rg8	l1	5,969.76	44.97	2.56

Recorded measurement scope: RTX 5070 Ti Laptop, CUDA 12.8; 32 x 2048-square pages; W=65,536 blocks; 256 intervals; batch=32; seed=756; dt=1024 and gain=16384 Q16; hops=2; mutation/jitter on. Three rotated sequential trials after one excluded warmup per variant. Clocks not locked. Wall time excludes allocation, initialization, snapshots and report output.

Three-trial cached/balanced ranges: BC5 39.135-39.219 ms; RG8 45.721-45.912 ms. These are observed ranges, not confidence intervals. The shared and uncached texture variants are slower than the reference in this workload.

Read the stopwatch correctly

updates/s = 16*visited_blocks / circulation_seconds
mean interval time = circulation_seconds / intervals
equivalent sweep time = stored_pairs / updates_per_second

A circulation timer excludes allocation/initialization/export only when the measured harness says so. Whole-process timing includes startup. A derived equivalent sweep time uses average throughput; it is not a directly timestamped visit to every block. Faster results for a different W, codec or numerical binding are different workloads.

ELI5, cached BC5: each interval revises 1,048,576 items in an average 0.153 ms. Revising the 134.22 million stored items once takes a derived 19.58 ms; the whole process median is 233.5 ms including startup.

Counters answer a different question

Saved profiler observation	Recorded scope/result
Cached balanced	Median evolve 91.94 us; L1/TEX hit 72.38%; active warps 59.27%; DRAM/peak 12.43%.
Cached L1 preference	L1/TEX hit 85.47%; evolve 96.93 us. Hits aggregate all evolve traffic, not just the LUT.
Profiler scope	BC5, same 32-page field and W; three evolve launches after eight skipped, no graph; replay counters, unlocked clocks and uncontrolled caches. Not the circulation benchmark.

The cache hitRatio setting is a preference, not an observed hit percentage. More cache hits do not necessarily imply a faster kernel. Occupancy, working-set size and arithmetic cost must be interpreted alongside unprofiled repeated runtime measurements.

Capacity and validation: what is actually established

Packed layout	Token bytes / 16 pairs	B/A bytes	Total bytes/pair
BC5	16	8	1.5
RG8	32	8	2.5

BC5 uses 40% fewer payload bytes than RG8 with the same exact B/A grouping. Neither the extra channels nor the second operator bank are free. A complete field sweep has work at least proportional to the payload block count.

Codec / variant	Payload GiB	Stored pairs, million	Free MiB / sweeps
bc5 / cached / balanced	9.891	7,079.99	892.0 / 1
rg8 / cached / balanced	9.883	4,244.64	900.0 / 2

Fresh automatic allocation runs, fill=0.92 with at least 768 MiB reserved; page side4096, W=65,536, 8,192 intervals. Both initialized and swept. This is exercised large-field capacity with desktop headroom, not maximum allocation or DRAM bandwidth saturation.

Saved acceptance summary

Evidence	Result and scope
Complete snapshots	46 passing equality/rejection checks across four variants, two codecs, partial windows, graph/sequential and continuation; includes v3 rejection.
CPU numerical suites	Math 1,483,985 assertions; core 296,554; D4 cache 1,253,890; predicates 187,164 exact comparisons + 3,598 coverage assertions.
GPU component checks	All 65,536 D4 pairs exactly match direct GPU charts; all 3,840 live support bits match CPU signed SDF; complete 64-byte stages and all 31 mutated records match CPU in self-tests.
Sanitizers	8 passing runs: memcheck, racecheck, synccheck, initcheck for both codecs; 3 pages, W=37, 17 intervals, 5 hops.
Decoder scope	RG8 tokens exact; native BC5 differs from software palette by at most one token in tested reads. CPU stage oracle starts with actual native decoded bytes.
Quantum component scope	H and phase interference tested directly; LP8 quotient loss measured separately. No entangled-state model or full-recurrence quantum fidelity claim.
Evidence location	results/integer/validation_fast_masks/validation.json; independent tests and sanitizer logs retained.

The validation harness ran 70 processes for its 46 checks. Separate regression runs passed all 12 CTest entries and 27 Python tests. Those totals include historical/regression checks and are not additional independent proofs of quantum behavior.

Exact snapshots are stronger evidence than matching only an operator digest. State equality applies to the tested inputs and decoder law. Native BC5 rounding is a separately checked boundary; a tolerated local decode unit cannot excuse arbitrary later state divergence. Integer-1 snapshots use version 4 and reject historical floating version 3.

Reproduce, inspect and extend

Use a Visual Studio 2022 developer shell with CUDA 12.8 or later. The following uses a separate Ninja Release build folder; PowerShell continuations are real backticks. The integer CLI accepts integer dt/gain words.

```
cmake -S . -B build-integer -G Ninja `
  -DCMAKE_BUILD_TYPE=Release
cmake --build build-integer --parallel
.\build-integer\dawnwood_integer_cached.exe --self-test
.\build-integer\dawnwood_integer_cached.exe --codec bc5 `
  --pages 32 --page-side 2048 --blocks-per-step 65536 `
  --dt-q16 1024 --inverse-gain-q16 16384 `
  --steps 256 --batch 32 --report work\integer.json
python tools\build_integer_pdf.py
```

The recommended cached executable defaults to balanced cache policy. Adding --cache-policy 11 improved the recorded large-case circulation median; profile counter results are scoped separately. Keep the numerical configuration fixed when comparing policies.

The saved final summary and individual reports identify variant names, exact commands and hashes. Repeat with the matching integer reference and the same parameters. A comparison to the earlier floating model is a cross-model comparison, not bitwise optimization evidence.

Project-relative artifact	Why it matters
model/integer_binding.json	Numerical units, rounding, model/snapshot identifiers and evidence contract.
docs/INTEGER_REQUIREMENTS.md	Literal source requirements and a complete code-plus-evidence checklist.
results/integer/theory_diagnostics.json	Phase-sweep plot data, error distributions, invariants and compiler/source/binary provenance.
results/integer/quantum_mechanism_checks.json	Actual H-squared and H-phase-H probes, normalized overlap, unnormalized norm ratios and phase-sweep error.
results/integer/final_summary.json	New performance, capacity, validation and instruction-audit summary consumed by this report.
tools/verify_integer_device_code.py	PTX/SASS checks and explicit native sampler / constant-only HFMA2 exceptions.

Primary references

Source: Supplied double-slit-theory.pdf, 24 pages; unchanged SHA-256 e3cf7d49e6a4942c7ccad4805f6a2a07e1c817753ff8d384b1f2e37791ebbe27.

NVIDIA: [Texture object API](#); [L2 cache control](#); [PTX ISA](#); [Binary utilities](#).

Physics: [Feynman Lectures III.1](#): amplitude addition, interference and probability of detection.

Packing: [Microsoft block-compression reference](#): BC5 two-channel 16-byte blocks.

No physical experiment, quantum hardware, universal-computation proof or beneficial-mutation theorem is implied. The delivered object is an explicit, editable, finite classical recurrence whose numerical and hardware properties can be measured.